

Nithin Raghava Aitha

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SUMMARY

Software Engineer with 4+ years building scalable payment systems and ML-powered backend solutions. Expertise in Python, FastAPI, Kubernetes, and cloud infrastructure (AWS/Azure) with proven experience in fraud detection, PCI-DSS compliance, and high-throughput data processing. Delivered measurable impact reducing settlement times by 75%, API latency by 65%, and processing costs through intelligent auto-scaling and automation.

SKILLS

Programming Languages: Python, SQL

Frameworks & Libraries: FastAPI, PyTorch, TensorFlow, LangChain, Apache Spark, BERT, spaCy, MLflow

Databases & Caching: PostgreSQL, Redis

Cloud Platforms: AWS (ECS, Lambda, API Gateway, X-Ray, WAF), Azure (AKS, Azure ML, Monitor, Application Insights, Cognitive Services)

DevOps & Infrastructure: Kubernetes, Docker, GitHub Actions, Jenkins, CI/CD Pipelines, Blue/Green Deployments

APIs & Integration: REST APIs, GraphQL, OAuth 2.0, Payment Gateway Integrations

Machine Learning & AI: Model Deployment, MLflow, Fraud Detection, NLP, GPT-4 Integration, Model Lifecycle Management

Domain Expertise: Payment Processing, PCI-DSS Compliance, Multi-Tenant Architecture, Enterprise Backup Systems

EXPERIENCE

Global Payments, USA | Software Engineer

Sep 2024 - Present

- Developed core components of a distributed payment reconciliation service handling multi-channel merchant settlements using Python, FastAPI, PostgreSQL, and Redis on AWS ECS - contributed to reducing settlement cycle time from 4 hours to under 1 hour while maintaining 99.9% uptime.
- Built secure integrations with payment gateways and card networks implementing OAuth 2.0 authentication, AWS API Gateway rate limiting, and WAF security controls that supported the team's PCI-DSS Level 1 compliance certification.
- Optimized performance-critical sections of the merchant analytics API by identifying GraphQL N+1 query patterns through AWS X-Ray tracing and refactoring data aggregation logic, reducing P99 response times by 65% for thousands of active merchants.
- Implemented CI/CD pipeline improvements migrating payment microservices from Jenkins to GitHub Actions with Kubernetes blue/green deployment patterns, enabling zero-downtime production releases.
- Developed real-time fraud detection features integrating PyTorch ML models into the authorization pipeline via AWS Lambda to score card-not-present transactions and flag cross-border fraud patterns.
- Created an AI-powered payment exception resolution tool leveraging LangChain and GPT-4 to automate dispute classification and chargeback documentation workflows, reducing manual processing time by 60% for routine cases.

Freshworks Inc, India | Software Engineer

Jan 2021 - July 2023

- Built a high-throughput data processing backend using Python, FastAPI, and Apache Spark on Azure Kubernetes Service to power enterprise backup workflows, processing terabytes of client data daily with failure rates under 2%.
- Contributed to a multi-tenant backend platform architecture enabling fully isolated compute environments per enterprise client, which reduced new-client onboarding time by 50% and improved team scalability.
- Developed a model lifecycle management service on Azure ML using MLflow and REST APIs to automate experiment tracking and model promotion workflows, compressing data science release cycles from weeks to under one week.
- Implemented auto-scaling infrastructure on Azure Kubernetes Service using workload forecasting from Azure Monitor metrics, eliminating over-provisioning and reducing cloud costs in the first production quarter.
- Built backend services for document intelligence automation integrating BERT, spaCy, and Azure Cognitive Services via REST APIs, reducing manual processing overhead across multiple product teams.
- Designed and implemented RESTful APIs for backup job orchestration and metadata retrieval used by frontend and mobile teams, handling 50K+ requests per day with P95 latency under 200ms through PostgreSQL query optimization and Redis caching.
- Established comprehensive monitoring and alerting infrastructure using Azure Monitor, Application Insights, and custom Python logging frameworks, reducing mean time to detection for production incidents by 40% and improving on-call response effectiveness.

CERTIFICATION

AWS Certified Machine Learning - Specialty.
TensorFlow Developer Certificate.

EDUCATION

Master of Science in Computer Science, George Washington University, Washington, D.C

Bachelor of Science in Computer Science & Engineering, BVRIT, India